



A Reflection on Artificial Intelligence and Server-Side Technology

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The development of our final project, Corte, a bus navigation system, required us to quickly learn several new technical concepts. Artificial intelligence supported our learning by explaining difficult topics and helping us improve certain parts of our code. Our PHP backend needed to follow a strict and reliable structure. AI gave us ideas, but we still had to design the system ourselves. We used a three-tier architecture:

1. API endpoints to receive requests
2. Business logic in the BusRouteFinder class to handle route calculations
3. A data access layer to communicate with the database

Security requirements also guided how we built the server. We implemented prepared statements to prevent SQL injections, applied password hashing with proper salts, handled PHP session tokens correctly, and configured Apache logs to track errors and activity. AI reminded us of OWASP recommendations, but we had to understand and apply them based on our project's specific risks. For example, we had to design the system carefully to ensure that the user's GPS coordinates were not accidentally exposed through the API. This became more difficult when our authentication system expanded from simple user login to admin-level access controls.

AI helped us research routing algorithms by summarising which approaches were suitable for public transit systems. Even so, we wrote the distance-calculation logic ourselves because AI-generated implementations often contained mistakes or did not match our performance needs. We also manually optimised parts of the algorithm to prevent slow response times when handling large route datasets.

Designing RESTful endpoints required human decision-making as well. We had to handle multiple cases, such as successful route calculations, invalid latitude–longitude pairs, and missing datasets. AI-generated example JSON structures, but we eventually refined the format through repeated testing with actual input from our navigation system.

The most helpful contribution from AI was its ability to quickly summarise research sources and generate the Overpass API commands we used to extract bus stop and road data from OpenStreetMap. Even with this support, the core of the server-side work depended on our understanding, planning, and debugging.

In the end, AI assisted us, but it did not replace the hands-on learning involved in building a secure and efficient server. Most of the critical decisions—especially around security, routing accuracy, and API design—still require human judgment and careful implementation.